

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Personal Portfolio Website

Version 1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for a futuristic, interactive, and visually immersive personal portfolio website. The platform will serve as a professional digital presence showcasing the portfolio owner's expertise in Artificial Intelligence, Machine Learning, Full-Stack Web Development, Network Engineering, Cybersecurity, and Data Analytics.

The website will provide visitors, recruiters, clients, researchers, and collaborators with an engaging experience through advanced 3D visualizations, interactive animations, project showcases, and professional information.

1.2 Scope

The system will function as a modern portfolio platform featuring:

- Interactive 3D user interface
- Professional profile presentation
- Dynamic project showcase
- Technical skills visualization
- Experience and achievements section
- Research and publications section
- Downloadable CV/Resume
- Contact and collaboration portal
- Responsive design for desktop and mobile devices
- High-performance rendering and animations

The website shall emphasize innovation, technical excellence, and modern user experience principles.

1.3 Intended Audience

The platform is intended for:

- Recruiters
 - Hiring Managers
 - Research Supervisors
 - Potential Clients
 - Startup Founders
 - Academic Institutions
 - Industry Professionals
 - Technology Enthusiasts
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2. Overall Description

2.1 Product Perspective

The website will act as a digital professional identity and personal brand platform.

The design philosophy combines:

- Futuristic UI
 - Cyberpunk-inspired visuals
 - AI-themed animations
 - Interactive 3D experiences
 - Professional portfolio functionality
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2.2 User Classes

Visitor

Can:

- Browse website
- View projects
- Read about skills
- Download CV
- Submit contact form

Recruiter

Can:

- Review technical skills
- Access projects
- Evaluate experience
- Download resume
- Contact owner

Administrator (Portfolio Owner)

Can:

- Manage projects
 - Update content
 - Manage blog posts
 - View contact submissions
 - Manage resume versions
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3. Functional Requirements

3.1 Landing Experience

FR-001

The system shall display a full-screen futuristic landing page.

FR-002

The system shall render a 3D animated environment upon page load.

FR-003

The system shall display the portfolio owner's professional title:

"AI Engineer | Full-Stack Developer | Network Engineer"

FR-004

The system shall support smooth scrolling navigation.

FR-005

The system shall provide a call-to-action button directing visitors to projects.

3.2 3D Interactive Scene

FR-006

The system shall display a 3D scene built using WebGL technologies.

FR-007

The scene shall respond to user mouse movement.

FR-008

The scene shall support camera transitions.

FR-009

The scene shall maintain at least 60 FPS on supported devices.

FR-010

The system shall include animated AI, networking, and cybersecurity themed visual elements.

Examples:

- Neural network visualization
- Data streams
- Floating network nodes
- Holographic interfaces
- Cyber grids

3.3 About Section

FR-011

The system shall display professional biography.

FR-012

The system shall present educational background.

FR-013

The system shall display career objectives.

FR-014

The system shall include an animated timeline.

3.4 Skills Section

FR-015

The system shall display skills grouped by domain.

Domains:

- Artificial Intelligence
- Machine Learning
- Web Development
- Network Engineering
- Cybersecurity
- Data Analytics

FR-016

Skills shall be displayed as interactive 3D elements.

FR-017

Users shall view additional information when hovering over skills.

3.5 Project Showcase

FR-018

The system shall display featured projects.

FR-019

Each project shall include:

- Title
- Description
- Technologies
- Screenshots
- GitHub Repository
- Live Demo

FR-020

Projects shall open within immersive modal windows.

FR-021

The system shall support project filtering.

Filters:

- AI
 - Networking
 - Cybersecurity
 - Web Development
 - Data Analysis
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3.6 Research Section

FR-022

The system shall display research interests.

FR-023

The system shall present publications and technical reports.

FR-024

The system shall support downloadable PDFs.

3.7 Resume Section

FR-025

The system shall provide CV download functionality.

FR-026

The system shall support multiple resume versions.

Examples:

- AI Resume
- Software Resume

3.8 Contact System

FR-027

The system shall provide a contact form.

Fields:

- Name
- Email
- Subject
- Message

FR-028

The system shall validate inputs.

FR-029

The system shall notify the administrator upon submission.

3.9 Blog Section

FR-030

The system shall provide a technical blog.

FR-031

The administrator shall create and edit articles.

FR-032

Visitors shall search articles.

4. Non-Functional Requirements

4.1 Performance

NFR-001

Initial page load shall be less than 4 seconds.

NFR-002

Animations shall maintain smooth performance.

NFR-003

3D rendering shall degrade gracefully on low-end devices.

4.2 Security

NFR-004

The system shall use HTTPS.

NFR-005

Contact forms shall implement CSRF protection.

NFR-006

User input shall be sanitized.

NFR-007

Rate limiting shall be implemented.

4.3 Usability

NFR-008

The interface shall be intuitive.

NFR-009

The website shall support keyboard navigation.

NFR-010

The website shall meet accessibility standards.

4.4 Compatibility

NFR-011

The website shall support:

- Chrome
- Firefox
- Edge
- Safari

NFR-012

The website shall be fully responsive.

5. Technology Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

3D Engine

- Three.js
- React Three Fiber
- Drei

Animation

- Framer Motion
- GSAP

Backend

- Django
- Django REST Framework

Database

- PostgreSQL

Deployment

- Docker
 - Nginx
 - Linux VPS
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6. Future Enhancements

- AI-powered portfolio assistant
 - Interactive chatbot
 - Voice navigation
 - Real-time visitor analytics
 - AI resume reviewer
 - AI project recommendation engine
 - WebXR/VR support
 - Interactive 3D virtual office
 - Digital twin workspace
 - Live coding demonstrations
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7. Success Criteria

The project shall be considered successful when:

- The website provides a unique futuristic experience.
- Visitors can easily explore projects and skills.
- Recruiters can assess qualifications quickly.
- The platform performs smoothly across devices.
- The website effectively represents the owner's professional brand.